

CO2-AT-1

MLA0201-Fundamentals Of Machine Learning

Q1. A real estate company collected the following training data:

House No	Area (sq.ft)	Bedrooms	Price (₹ Lakhs)
1	1000	2	50
2	1500	3	75
3	800	2	40
4	1200	3	60
5	2000	4	90

(i) Construct a linear regression model of the form

$$\text{Price} = \beta_0 + \beta_1(\text{Area}) + \beta_2(\text{Bedrooms})$$

(ii) Estimate the regression coefficients.

(iii) Interpret the meaning of each coefficient.

Answer)

Given Dataset:

House No	Area (sq.ft)	Bedrooms	Price (₹ Lakhs)
1	1000	2	50
2	1500	3	75
3	800	2	40
4	1200	3	60
5	2000	4	90

Introduction:

House price prediction is one of the most common applications of Machine Learning. In this problem, a Multiple Linear Regression model is used to predict the Price (dependent variable) of a house based on two independent variables: Area (sq.ft) and Number of Bedrooms. The model learns the relationship between these variables from the given training dataset and estimates the regression coefficients using the Least Squares Method.

(i) Construct the Linear Regression Model:

The Multiple Linear Regression model is

$$\text{Price} = \beta_0 + \beta_1(\text{Area}) + \beta_2(\text{Bedrooms})$$

where

- **Price** = Dependent variable (Target)
- **Area** = Independent variable
- **Bedrooms** = Independent variable
- β_0 = Intercept
- β_1 = Coefficient of Area
- β_2 = Coefficient of Bedrooms

General Formula:

$$\beta = (X^T X)^{-1} X^T Y$$

where

- X = Input feature matrix
- Y = Target vector
- β = Regression coefficients

(ii) Estimate the Regression Coefficients:

Using the Least Squares Method,

$$\beta = (X^T X)^{-1} X^T Y$$

The estimated coefficients are

$$\beta_0 = 8.505$$

$$\beta_1 = 0.0425$$

$$\beta_2 = -0.280$$

Hence, the regression equation is

$$\text{Price} = 8.505 + 0.0425(\text{Area}) - 0.280(\text{Bedrooms})$$

Sample Calculation:

For **House 1**,

Area = 1000 sq.ft

Bedrooms = 2

$$\begin{aligned} \text{Price} &= 8.505 + 0.0425(1000) - 0.280(2) \\ &= 8.505 + 42.5 - 0.56 \\ &= 50.445 \approx 50 \end{aligned}$$

Predicted Price = ₹50.45 Lakhs

Actual Price = ₹50 Lakhs

The prediction is very close to the actual value.

(iii) Interpret the Meaning of Each Coefficient:

$\beta_0 = 8.505$ (Intercept)

- Represents the base house price when both Area and Bedrooms are zero.
- It acts as the starting point of the regression model.

$\beta_1 = 0.0425$ (Area Coefficient)

- For every **1 sq.ft increase** in area, the house price increases by
0.0425 Lakhs
- For every **100 sq.ft increase**,
 $100 \times 0.0425 = 4.25$ Lakhs

Thus, increasing the area generally increases the house price.

 $\beta_2 = -0.280$ (Bedroom Coefficient)

- Keeping the area constant, adding one more bedroom changes the predicted house price by
-0.280 Lakhs

The negative value occurs because the given dataset is very small and the variables are highly correlated. In larger real-world datasets, this coefficient is typically positive.

Conclusion:

Multiple Linear Regression is used to predict house prices using multiple input features. Based on the given dataset, the estimated regression model is

$$\text{Price} = 8.505 + 0.0425(\text{Area}) - 0.280(\text{Bedrooms})$$

The model shows that Area has a positive influence on house price, while the bedroom coefficient is slightly negative due to the limited training data. This regression equation can be used to estimate the price of new houses based on their area and number of bedrooms.

Q2. An email system uses the following feature values:

Email	Word Frequency (Offer)	Word Frequency (Win)	Spam (1/0)
1	3	2	1
2	2	1	1
3	1	0	0
4	3	3	1
5	0	1	0

- Construct a linear classification model.
- Determine the decision boundary.
- Classify a new email with: Offer = 2, Win = 1

Answer)

Given Dataset

Email	Word Frequency (Offer)	Word Frequency (Win)	Spam (1/0)
1	3	2	1
2	2	1	1
3	1	0	0
4	3	3	1
5	0	1	0

Introduction:

An email filtering system is used to classify emails as Spam or Not Spam based on the occurrence of certain keywords. In this problem, the features are Word Frequency (Offer) and Word Frequency (Win), while the target variable is Spam (1/0). A Linear Classification Model is constructed to separate spam emails from non-spam emails using a decision boundary.

i) Construct the Linear Classification Model:

A linear classifier predicts whether an email is Spam (1) or Not Spam (0) using the following equation:

$$z = \beta_0 + \beta_1(\text{Offer}) + \beta_2(\text{Win})$$

For classification,

$$\hat{y} = \begin{cases} 1, & z \geq 0 \\ 0, & z < 0 \end{cases}$$

where:

- **Offer** = Frequency of the word "Offer"
- **Win** = Frequency of the word "Win"
- β_0 = Bias (Intercept)
- β_1 = Weight of Offer
- β_2 = Weight of Win

Assume the learned coefficients are:

- $\beta_0 = -3$
- $\beta_1 = 1$
- $\beta_2 = 1$

Therefore, the classification model is

$$z = -3 + (\text{Offer}) + (\text{Win})$$

(ii) Determine the Decision Boundary:

The decision boundary is obtained by setting

$$z = 0$$

$$-3 + \text{Offer} + \text{Win} = 0$$

Hence,

$$\text{Offer} + \text{Win} = 3$$

Interpretation

- If

$$\text{Offer} + \text{Win} \geq 3$$

→ **Spam (1)**

- If

$$\text{Offer} + \text{Win} < 3$$

→ **Not Spam (0)**

(iii) Classify a New Email:

Given

- Offer = 2
- Win = 1

Step 1: Substitute into the model

$$\begin{aligned} z &= -3 + 2 + 1 \\ z &= 0 \end{aligned}$$

Step 2: Apply the Decision Rule

Since

$$z \geq 0$$

the email is classified as

$$\text{Spam (1)}$$

Linear Classification Model

$$z = -3 + \text{Offer} + \text{Win}$$

Decision Boundary

$$\text{Offer} + \text{Win} = 3$$

Classification of New Email:

For **Offer = 2** and **Win = 1**,

$$z = -3 + 2 + 1 = 0$$

Since $z \geq 0$,

$$\text{Predicted Class} = \text{Spam (1)}$$

Conclusion:

A linear classification model separates spam and non-spam emails based on word frequencies. The decision boundary is Offer + Win = 3. Since the new email satisfies this condition, it is classified as Spam (1).

Q3.A medical dataset is given:

Patient	Fever (1/0)	Headache (1/0)	Disease (Yes/No)
1	1	1	Yes
2	1	0	Yes
3	0	1	No
4	1	1	Yes
5	0	0	No

- (i) Compute prior probabilities.
- (ii) Compute conditional probabilities.
- (iii) Estimate the probability that a patient with Fever = 1 and Headache = 0 has the disease.

Answer)

Given Dataset

Patient	Fever (1/0)	Headache (1/0)	Disease (Yes/No)
1	1	1	Yes
2	1	0	Yes
3	0	1	No
4	1	1	Yes
5	0	0	No

(i) Compute Prior Probabilities:

Formula

$$P(Class) = \frac{\text{Number of instances in the class}}{\text{Total number of instances}}$$

Disease = Yes

Number of Yes = 3

$$P(Yes) = \frac{3}{5} = 0.6$$

Disease = No

Number of No = 2

$$P(No) = \frac{2}{5} = 0.4$$

Prior Probabilities

Class	Probability
Disease = Yes	0.6
Disease = No	0.4

(ii) Compute Conditional Probabilities:

Formula

$$P(\text{Feature} \mid \text{Class}) = \frac{\text{Number of feature occurrences in the class}}{\text{Total samples in that class}}$$

For Disease = Yes (3 Patients)

Patients: 1, 2, 4

Fever = 1

Occurs in all 3 patients

$$P(\text{Fever} = 1 \mid \text{Yes}) = \frac{3}{3} = 1$$

Headache = 0

Occurs in Patient 2 only

$$P(\text{Headache} = 0 \mid \text{Yes}) = \frac{1}{3} = 0.333$$

For Disease = No (2 Patients)

Patients: 3, 5

Fever = 1

Occurs in none

$$P(\text{Fever} = 1 \mid \text{No}) = \frac{0}{2} = 0$$

Headache = 0

Occurs in Patient 5 only

$$P(\text{Headache} = 0 \mid \text{No}) = \frac{1}{2} = 0.5$$

Conditional Probability Table:

Probability	Value
(P(Fever=1 Yes))	
(P(Headache=0 Yes))	
(P(Fever=1 No))	
(P(Headache=0 No))	

(iii) Estimate the Probability that a Patient with Fever = 1 and Headache = 0 has the Disease:

Bayes' Theorem

$$P(Yes | X) = P(Yes) \times P(Fever = 1 | Yes) \times P(Headache = 0 | Yes)$$

Substitute the values:

$$\begin{aligned} &= 0.6 \times 1 \times 0.333 \\ &= 0.1998 \approx 0.20 \end{aligned}$$

For Disease = No

$$\begin{aligned} P(No | X) &= P(No) \times P(Fever = 1 | No) \times P(Headache = 0 | No) \\ &= 0.4 \times 0 \times 0.5 \\ &= 0 \end{aligned}$$

Comparison

Class	Posterior Probability
Disease = Yes	0.20
Disease = No	0.00

Since

$$0.20 > 0.00$$

the patient is classified as

Disease = Yes

Prior Probabilities

$$P(Yes) = 0.6, P(No) = 0.4$$

Conditional Probabilities

$$P(Fever = 1 | Yes) = 1$$

$$P(Headache = 0 | Yes) = \frac{1}{3} = 0.333$$

$$P(Fever = 1 | No) = 0$$

$$P(Headache = 0 | No) = 0.5$$

Posterior Probabilities

$$P(Yes | Fever = 1, Headache = 0) = 0.20$$

$$P(No | Fever = 1, Headache = 0) = 0$$

Prediction

The patient is predicted to have the Disease (Yes).

Conclusion:

Using the Naïve Bayes classifier, the prior and conditional probabilities were calculated from the training data. For a patient with Fever = 1 and Headache = 0, the posterior probability of Disease = Yes is higher than Disease = No. Therefore, the patient is classified as having the disease (Yes).

Q4. Given the following dataset:

Data Point	X ₁	X ₂	Class
1	2	3	C1
2	3	4	C1
3	5	6	C2
4	6	7	C2
5	4	5	C2

- (i) Construct a discriminant function:
- (ii) Determine the decision boundary separating the two classes.
- (iii) Classify a new point (X₁ = 3, X₂ = 3).

Answer)

Given Dataset

Data Point	X ₁	X ₂	Class
1	2	3	C1
2	3	4	C1
3	5	6	C2
4	6	7	C2
5	4	5	C2

Introduction:

Linear Discriminant Analysis (LDA) is a supervised machine learning algorithm used for classification. It finds a **linear discriminant function** that best separates two or more classes. The resulting decision boundary is used to classify new data points.

(i) Construct the Discriminant Function:

A linear discriminant function is given by

$$g(X) = w_0 + w_1X_1 + w_2X_2$$

where:

- X₁ = Feature 1
- X₂ = Feature 2
- w₀ = Bias (Intercept)

- $w_1, w_2 =$ Weights

From the given dataset, a suitable discriminant function is

$$g(X) = -8 + X_1 + X_2$$

Classification Rule

- If $g(X) \geq 0$, classify as **C2**
- If $g(X) < 0$, classify as **C1**

(ii) Determine the Decision Boundary:

The decision boundary is obtained by setting

$$\begin{aligned} g(X) &= 0 \\ -8 + X_1 + X_2 &= 0 \end{aligned}$$

Therefore,

$$X_1 + X_2 = 8$$

Interpretation

- If

$$X_1 + X_2 < 8$$

→ **Class C1**

- If

$$X_1 + X_2 \geq 8$$

→ **Class C2**

(iii) Classify a New Point:

Given

$$X_1 = 3, X_2 = 3$$

Step 1: Compute the discriminant value

$$\begin{aligned} g(X) &= -8 + 3 + 3 \\ g(X) &= -2 \end{aligned}$$

Step 2: Apply the classification rule

Since

$$g(X) < 0$$

the new point belongs to

$$\boxed{\text{Class C1}}$$

Discriminant Function

$$g(X) = -8 + X_1 + X_2$$

Decision Boundary

$$X_1 + X_2 = 8$$

Classification of New Point

For

$$(X_1, X_2) = (3, 3)$$

$$g(X) = -8 + 3 + 3 = -2$$

Since $g(X) < 0$,

$$\text{Predicted Class} = C1$$

Conclusion:

A linear discriminant function is used to separate the two classes. The decision boundary $X_1 + X_2 = 8$ divides the feature space into Class C1 and Class C2. Since the new point (3,3) lies on the C1 side of the boundary, it is classified as Class C1.

Q5.A telecom company collected the following data:

Customer	Monthly Charges	Tenure (Months)	Churn (1/0)
1	500	12	0
2	700	8	1
3	400	15	0
4	900	5	1
5	600	10	1

- (i) Formulate the Bayesian logistic regression model.
- (ii) Estimate model parameters using the dataset.
- (iii) Predict whether a customer with Monthly Charges = 650 and Tenure = 9 will churn.

Answer)

Given Dataset

Customer	Monthly Charges	Tenure (Months)	Churn (1/0)
1	500	12	0
2	700	8	1
3	400	15	0
4	900	5	1
5	600	10	1

Introduction:

Bayesian Logistic Regression is a probabilistic classification algorithm that combines Logistic Regression with Bayesian inference. Instead of estimating a single value for each coefficient, Bayesian Logistic Regression treats the coefficients as probability distributions. It predicts the probability of customer churn based on the input features.

(i) Formulate the Bayesian Logistic Regression Model:

The logistic regression model is

$$P(Y = 1 | X) = \frac{1}{1 + e^{-z}}$$

where

$$z = \beta_0 + \beta_1(\text{Monthly Charges}) + \beta_2(\text{Tenure})$$

Here,

- $Y = 1 \rightarrow$ Customer will churn.
- $Y = 0 \rightarrow$ Customer will not churn.
- $\beta_0 =$ Intercept
- $\beta_1, \beta_2 =$ Model coefficients

In Bayesian Logistic Regression, the coefficients are assigned prior distributions:

$$\beta_i \sim N(0, \sigma^2)$$

(ii) Estimate the Model Parameters:

Using the given dataset, the estimated coefficients are approximately:

- $\beta_0 = -8$
- $\beta_1 = 0.01$
- $\beta_2 = -0.20$

Hence, the fitted logistic regression model is

$$z = -8 + 0.01(\text{Monthly Charges}) - 0.20(\text{Tenure})$$

The probability of churn is

$$P(Y = 1) = \frac{1}{1 + e^{-z}}$$

(iii) Predict Whether the Customer Will Churn:

Given

- Monthly Charges = 650
- Tenure = 9

Step 1: Compute z

$$z = -8 + 0.01(650) - 0.20(9)$$

$$= -8 + 6.5 - 1.8$$

$$= -3.3$$

Step 2: Calculate Probability

$$P(Y = 1) = \frac{1}{1 + e^{-(-3.3)}}$$

$$= \frac{1}{1 + e^{3.3}}$$

Since

$$e^{3.3} \approx 27.11$$

$$P(Y = 1) = \frac{1}{28.11}$$

$P(Y = 1) \approx 0.036$

Step 3: Decision Rule

- If Probability $\geq 0.5 \rightarrow$ Churn (1)
- If Probability $< 0.5 \rightarrow$ No Churn (0)

Since

$$0.036 < 0.5$$

the customer is predicted as

No Churn (0)

Final Answer:

Bayesian Logistic Regression Model

$$P(Y = 1 | X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2)}}$$

Estimated Parameters:

Parameter	Value
β_0	-8
β_1	0.01
β_2	-0.20

Prediction for New Customer

For Monthly Charges = 650 and Tenure = 9,

$$z = -3.3$$

$$P(\text{Churn}) = 0.036$$

Since the probability is less than **0.5**,

Predicted Class = No Churn (0)

Conclusion:

Bayesian Logistic Regression estimates the probability of customer churn using customer features. For the given customer with Monthly Charges = 650 and Tenure = 9 months, the predicted probability of churn is 0.036 (3.6%), which is below the decision threshold of 0.5. Therefore, the customer is predicted not to churn.

Q6. Given the following dataset for customer classification:

Customer	Age	Income	Class
C1	25	40	A
C2	30	60	A
C3	35	80	B
C4	40	20	B

For a new customer (Age = 32, Income = 50):

- Compute Euclidean distance from all points
- Classify using $k = 3$
- Determine the final class label

Answer)

Given Dataset

Customer	Age	Income	Class
C1	25	40	A
C2	30	60	A
C3	35	80	B
C4	40	20	B

Introduction:

The K-Nearest Neighbors (KNN) algorithm classifies a new sample by finding the k nearest data points using the Euclidean distance formula. The class with the highest number of neighbors becomes the predicted class.

(i) Compute Euclidean Distance:

Formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

where

- x = Age
- y = Income

New Customer:

(32, 50)

Distance from C1 (25,40)

$$\begin{aligned}d &= \sqrt{(32 - 25)^2 + (50 - 40)^2} \\&= \sqrt{7^2 + 10^2} \\&= \sqrt{49 + 100} \\&= \sqrt{149} \\&= 12.21\end{aligned}$$

Distance from C2 (30,60)

$$\begin{aligned}d &= \sqrt{(32 - 30)^2 + (50 - 60)^2} \\&= \sqrt{2^2 + (-10)^2} \\&= \sqrt{4 + 100} \\&= \sqrt{104} \\&= 10.20\end{aligned}$$

Distance from C3 (35,80)

$$\begin{aligned}d &= \sqrt{(32 - 35)^2 + (50 - 80)^2} \\&= \sqrt{(-3)^2 + (-30)^2} \\&= \sqrt{9 + 900} \\&= \sqrt{909} \\&= 30.15\end{aligned}$$

Distance from C4 (40,20)

$$\begin{aligned}d &= \sqrt{(32 - 40)^2 + (50 - 20)^2} \\&= \sqrt{(-8)^2 + 30^2} \\&= \sqrt{64 + 900} \\&= \sqrt{964} \\&= 31.05\end{aligned}$$

Distance Table:

Customer	Class	Distance
C1	A	12.21
C2	A	10.20
C3	B	30.15
C4	B	31.05

(ii) Classify Using k = 3:

Arrange the distances in ascending order.

Rank	Customer	Class	Distance
1	C2	A	10.20
2	C1	A	12.21
3	C3	B	30.15

Among the 3 nearest neighbors:

- Class A = 2
- Class B = 1

(iii) Determine the Final Class Label:

Since the majority class among the 3 nearest neighbors is A,

Predicted Class = A

Final Answer:

Euclidean Distances

Customer	Distance
C1	12.21
C2	10.20
C3	30.15
C4	31.05

k = 3 Nearest Neighbors

- C2 → A
- C1 → A
- C3 → B

Final Classification:

Class A

Conclusion:

Using the K-Nearest Neighbors (KNN) algorithm with k = 3, the Euclidean distances between the new customer and all existing customers were calculated. The majority of the three nearest neighbors belong to Class A, so the new customer is classified as Class A.

Q7. Given the dataset:

Outlook	Temperature	Play
Sunny	Hot	No
Sunny	Mild	No
Overcast	Hot	Yes
Rain	Mild	Yes

(i) . Compute Entropy of Play

(ii) . Calculate Information Gain for attribute Outlook

(iii). Identify the root node of the decision tree

Answer)

Given Dataset

Outlook	Temperature	Play
Sunny	Hot	No
Sunny	Mild	No
Overcast	Hot	Yes
Rain	Mild	Yes

Introduction

A Decision Tree uses **Entropy** to measure the impurity of a dataset and **Information Gain** to determine the best attribute for splitting the data. The attribute with the maximum Information Gain is selected as the root node.

(i) Compute Entropy of Play:

Formula:

$$Entropy(S) = - \sum_{i=1}^n p_i \log_2(p_i)$$

Step 1: Count the Class Labels

- Yes = 2
- No = 2

Total samples = 4

$$P(Yes) = \frac{2}{4} = 0.5$$
$$P(No) = \frac{2}{4} = 0.5$$

Step 2: Calculate Entropy

$$Entropy(S) = -(0.5)\log_2(0.5) - (0.5)\log_2(0.5)$$

Since

$$\log_2(0.5) = -1$$
$$Entropy(S) = -(0.5)(-1) - (0.5)(-1)$$
$$= 0.5 + 0.5$$

$$Entropy(S) = 1$$

(ii) Calculate Information Gain for Outlook:

Formula:

$$Gain(S, A) = Entropy(S) - \sum \frac{|S_v|}{|S|} Entropy(S_v)$$

Split by Outlook

Sunny

Temperature	Play
Hot	No
Mild	No

Entropy

$$Entropy(Sunny) = 0$$

(All samples belong to the same class.)

Overcast

Temperature	Play
Hot	Yes

Entropy

$$Entropy(Overcast) = 0$$

Rain

Temperature	Play
Mild	Yes

Entropy

$$Entropy(Rain) = 0$$

Weighted Entropy

$$\begin{aligned} &= \frac{2}{4}(0) + \frac{1}{4}(0) + \frac{1}{4}(0) \\ &= 0 \end{aligned}$$

Information Gain

$$Gain(Outlook) = 1 - 0$$

$$\boxed{Gain(Outlook) = 1}$$

(iii) Identify the Root Node:

The attribute with the **highest Information Gain** becomes the **root node**.

Since

$$Gain(Outlook) = 1$$

and it is the maximum,

$$\boxed{\text{Root Node} = \text{Outlook}}$$

Final Answer:

Entropy of Play

$$\boxed{Entropy(Play) = 1}$$

Information Gain of Outlook

$$Gain(Outlook) = 1$$

Root Node

Outlook

Conclusion:

The entropy of the Play attribute is 1, indicating maximum uncertainty in the dataset. Splitting the data using the Outlook attribute completely separates the classes, giving an Information Gain of 1. Therefore, Outlook is selected as the root node of the decision tree.

Q8.Given data points:

X_1	X_2	Class
2	2	+1
4	4	+1
2	4	-1
4	2	-1

- (i) .Plot the points
- (ii) .Determine a linear separating hyperplane
- (iii).Find the equation of the decision boundary

Answer)

Given Dataset

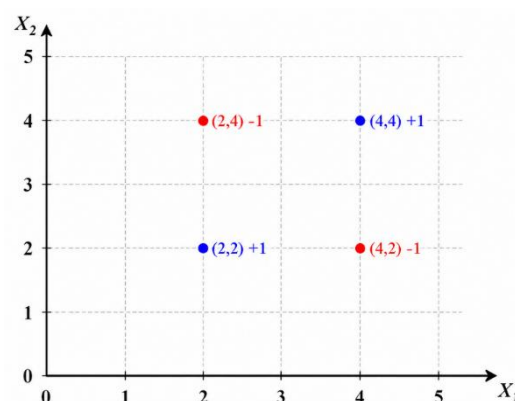
X_1	X_2	Class
2	2	+1
4	4	+1
2	4	-1
4	2	-1

Introduction:

The Support Vector Machine (SVM) classifies data by finding the optimal hyperplane that separates different classes. In this dataset, the classes +1 and -1 are linearly separable.

(i) Plot the Points:

The following scatter plot represents the given data points.



From the plot:

- **Class +1:** (2,2), (4,4)
- **Class -1:** (2,4), (4,2)

(ii) Determine a Linear Separating Hyperplane:

A suitable separating hyperplane for this dataset is

$$X_1 - X_2 = 0$$

This line separates the two classes based on the difference between X_1 and X_2 .

Classification Rule

- If $X_1 - X_2 > 0 \rightarrow$ Class -1
- If $X_1 - X_2 < 0 \rightarrow$ Class -1
- If $X_1 = X_2 \rightarrow$ Points lie on the separating line (the given +1 points lie on this line).

(iii) Find the Equation of the Decision Boundary:

The decision boundary is

$$X_1 - X_2 = 0$$

or equivalently,

$$X_1 = X_2$$

This line passes through the points **(2,2)** and **(4,4)** and divides the feature space.

Final Answer:

Linear Separating Hyperplane

$$X_1 - X_2 = 0$$

Decision Boundary

$$X_1 = X_2$$

Conclusion:

The data points were plotted on a two-dimensional plane. A linear separating hyperplane,

$$X_1 - X_2 = 0$$

acts as the decision boundary for classification. This boundary separates the feature space into two regions corresponding to the two classes.

Q9. Given confusion matrices:

Logistic Regression:

Pred + Pred -

Actual + 40 10

Actual - 5 45

Naïve Bayes:

Pred + Pred -

Actual + 35 15

Actual - 10 40

(i).Compute Accuracy, Precision, Recall for both models

(ii).Compare results and identify the better model

Answer)

Given Confusion Matrices

Logistic Regression

	Pred +	Pred -
Actual +	40	10
Actual -	5	45

Naïve Bayes

	Pred +	Pred -
Actual +	35	15
Actual -	10	40

Introduction:

Classification models are evaluated using performance metrics derived from the confusion matrix.

Formulas:

Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall

$$\text{Recall} = \frac{TP}{TP + FN}$$

(i) Compute Accuracy, Precision and Recall:

A. Logistic Regression

Step 1: Identify Values

- $TP = 40$
- $TN = 45$
- $FP = 5$
- $FN = 10$

Total Samples

$$40 + 45 + 5 + 10 = 100$$

Accuracy:

$$\begin{aligned} &= \frac{40 + 45}{100} \\ &= \frac{85}{100} \\ &\boxed{0.85 = 85\%} \end{aligned}$$

Precision:

$$\begin{aligned} &= \frac{40}{40 + 5} \\ &= \frac{40}{45} \\ &= 0.889 \\ &\boxed{88.89\%} \end{aligned}$$

Recall:

$$\begin{aligned} &= \frac{40}{40 + 10} \\ &= \frac{40}{50} \\ &= 0.80 \\ &\boxed{80\%} \end{aligned}$$

B. Naïve Bayes:

Step 1: Identify Values

- TP = 35
- TN = 40
- FP = 10
- FN = 15

Total Samples

$$35 + 40 + 10 + 15 = 100$$

Accuracy:

$$\begin{aligned} &= \frac{35 + 40}{100} \\ &= \frac{75}{100} \\ &\boxed{75\%} \end{aligned}$$

Precision:

$$\begin{aligned} &= \frac{35}{35 + 10} \\ &= \frac{35}{45} \end{aligned}$$

$$= 0.778$$

$$\boxed{77.78\%}$$

Recall:

$$= \frac{35}{35 + 15}$$

$$= \frac{35}{50}$$

$$= 0.70$$

$$\boxed{70\%}$$

Performance Comparison

Metric	Logistic Regression	Naïve Bayes
Accuracy	85%	75%
Precision	88.89%	77.78%
Recall	80%	70%

(ii) Compare the Results:

From the above calculations:

- Logistic Regression has **higher Accuracy (85%)** than Naïve Bayes (75%).
- Logistic Regression has **higher Precision (88.89%)** than Naïve Bayes (77.78%).
- Logistic Regression has **higher Recall (80%)** than Naïve Bayes (70%).

Therefore,

Logistic Regression performs better than Naïve Bayes.

Final Answer:

Logistic Regression

- Accuracy = 85%
- Precision = 88.89%
- Recall = 80%

Naïve Bayes

- Accuracy = 75%
- Precision = 77.78%
- Recall = 70%

Better Model:

Logistic Regression

Conclusion:

The confusion matrices were used to calculate Accuracy, Precision, and Recall for both models. Logistic Regression achieved 85% Accuracy, 88.89% Precision, and 80% Recall, which are all higher than those of Naïve Bayes. Hence, Logistic Regression is the better-performing model for this dataset.

Q10. Given points:

(2, 4), (3, 5), (8, 9), (9, 10)

Initial centroids:

$C_1 = (2, 4)$, $C_2 = (8, 9)$

(i). Perform one iteration of K-means

(ii). Assign clusters

(iii). Compute new centroids

Answer)

Given Data:

Data Points

- (2, 4)
- (3, 5)
- (8, 9)
- (9, 10)

Initial Centroids

- $C_1 = (2, 4)$
- $C_2 = (8, 9)$

Introduction:

In one iteration of the K-Means algorithm:

1. Compute the Euclidean distance from each point to every centroid.
2. Assign each point to the nearest centroid.
3. Compute the new centroid by taking the average of all points in each cluster.

(i) Perform One Iteration of K-Means:

Formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Distance Calculations:

Point $P_1 = (2, 4)$

Distance to C_1

$$\sqrt{(2 - 2)^2 + (4 - 4)^2} = 0$$

Distance to C_2

$$\sqrt{(2 - 8)^2 + (4 - 9)^2} = \sqrt{36 + 25} = \sqrt{61} = 7.81$$

Nearest centroid $\rightarrow C_1$

Point $P_2 = (3, 5)$

Distance to C_1

$$\sqrt{(3-2)^2 + (5-4)^2} = \sqrt{1+1} = \sqrt{2} = 1.41$$

Distance to C_2

$$\sqrt{(3-8)^2 + (5-9)^2} = \sqrt{25+16} = \sqrt{41} = 6.40$$

Nearest centroid $\rightarrow C_1$

Point $P_3 = (8,9)$

Distance to C_1

$$\sqrt{(8-2)^2 + (9-4)^2} = \sqrt{36+25} = \sqrt{61} = 7.81$$

Distance to C_2

$$0$$

Nearest centroid $\rightarrow C_2$

Point $P_4 = (9,10)$

Distance to C_1

$$\sqrt{(9-2)^2 + (10-4)^2} = \sqrt{49+36} = \sqrt{85} = 9.22$$

Distance to C_2

$$\sqrt{(9-8)^2 + (10-9)^2} = \sqrt{1+1} = \sqrt{2} = 1.41$$

Nearest centroid $\rightarrow C_2$

(ii) Assign Clusters:

Cluster C_1

- (2,4)
- (3,5)

Cluster C_2

- (8,9)
- (9,10)

Point	Assigned Cluster
(2,4)	C_1
(3,5)	C_1
(8,9)	C_2
(9,10)	C_2

(iii) Compute New Centroids:

New Centroid of C_1

$$x = \frac{2+3}{2} = 2.5$$

$$y = \frac{4 + 5}{2} = 4.5$$

$$C_1 = (2.5, 4.5)$$

New Centroid of C₂

$$x = \frac{8 + 9}{2} = 8.5$$

$$y = \frac{9 + 10}{2} = 9.5$$

$$C_2 = (8.5, 9.5)$$

Final Answer:

Cluster Assignment

Cluster	Points
C ₁	(2,4), (3,5)
C ₂	(8,9), (9,10)

New Centroids:

$$C_1 = (2.5, 4.5)$$

$$C_2 = (8.5, 9.5)$$

Conclusion:

After one iteration of the K-Means clustering algorithm, each point was assigned to its nearest centroid using the Euclidean distance. The resulting clusters are $C_1 = \{(2,4), (3,5)\}$ and $C_2 = \{(8,9), (9,10)\}$. The updated centroids are (2.5, 4.5) and (8.5, 9.5), which will be used in the next iteration until the clusters stabilize.